

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
# Load the data
```

```
df = pd.read_csv("International sale Report.csv.zip")
df
```

	index	DATE	Months		CUSTOMER	Style \
0	0	06-05-21	Jun-21	REVATHY	LOGANATHAN	MEN5004
1	1	06-05-21	Jun-21	REVATHY	LOGANATHAN	MEN5004
2	2	06-05-21	Jun-21	REVATHY	LOGANATHAN	MEN5004
3	3	06-05-21	Jun-21	REVATHY	LOGANATHAN	MEN5009
4	4	06-05-21	Jun-21	REVATHY	LOGANATHAN	MEN5011
...	...	...	...		...	...
37427	37427	AVIN	03-31-22		Mar-22	PJNE3423
37428	37428	AVIN	03-31-22		Mar-22	PJNE3404
37429	37429	AVIN	03-31-22		Mar-22	PJNE3423
37430	37430	AVIN	03-31-22		Mar-22	SET290
37431	37431	AVIN	03-31-22		Mar-22	SHIPPING

	SKU	Size	PCS	RATE	GROSS	AMT
0	MEN5004-KR-L	L	1.00	616.56	617.00	
1	MEN5004-KR-XL	XL	1.00	616.56	617.00	
2	MEN5004-KR-XXL	XXL	1.00	616.56	617.00	
3	MEN5009-KR-L	L	1.00	616.56	617.00	
4	MEN5011-KR-L	L	1.00	616.56	617.00	
...	...	...	...	...	...	...
37427	PJNE3423-KR-4XL	1.00	537.50	538.00	4.00	
37428	PJNE3404-KR-4XL	1.00	500.00	500.00	5.00	
37429	PJNE3423-KR-4XL	1.00	537.50	538.00	4.00	
37430	SET290-KR-DPT-M	1.00	812.50	812.00	7.00	
37431	SHIPPING	1.00	3666.00	3666.00	0.00	

```
[37432 rows x 10 columns]
```

```
# Data cleaning
```

```
df = df.drop_duplicates().dropna()
```

```
df['DATE'] = pd.to_datetime(df['DATE'], format="%m-%d-%y",
errors='coerce')
```

```
df['PCS'] = pd.to_numeric(df['PCS'], errors='coerce')
```

```
df['RATE'] = pd.to_numeric(df['RATE'], errors='coerce')
```

```
df['GROSS AMT'] = pd.to_numeric(df['GROSS AMT'], errors='coerce')
```

```
df = df.dropna() # Drop again after type conversion
```

```
df
```

	index	DATE	Months		CUSTOMER	Style \
0	0	2021-06-05	Jun-21	REVATHY	LOGANATHAN	MEN5004
1	1	2021-06-05	Jun-21	REVATHY	LOGANATHAN	MEN5004

2	2	2021-06-05	Jun-21	REVATHY	LOGANATHAN	MEN5004
3	3	2021-06-05	Jun-21	REVATHY	LOGANATHAN	MEN5009
4	4	2021-06-05	Jun-21	REVATHY	LOGANATHAN	MEN5011
...	...	...	...		...	...
17230	17230	2022-03-31	Mar-22		AVIN	SET276
17231	17231	2022-03-31	Mar-22		AVIN	PJNE3423
17232	17232	2022-03-31	Mar-22		AVIN	PJNE3404
17233	17233	2022-03-31	Mar-22		AVIN	PJNE3423
17234	17234	2022-03-31	Mar-22		AVIN	SET290

	SKU	Size	PCS	RATE	GROSS	AMT
0	MEN5004-KR-L	L	1.0	616.56		617.0
1	MEN5004-KR-XL	XL	1.0	616.56		617.0
2	MEN5004-KR-XXL	XXL	1.0	616.56		617.0
3	MEN5009-KR-L	L	1.0	616.56		617.0
4	MEN5011-KR-L	L	1.0	616.56		617.0
...	...	...	...	...	...	...
17230	SET276-KR-PP-L	L	1.0	1425.00		1425.0
17231	PJNE3423-KR-4XL	4XL	1.0	537.50		538.0
17232	PJNE3404-KR-4XL	4XL	1.0	500.00		500.0
17233	PJNE3423-KR-4XL	4XL	1.0	537.50		538.0
17234	SET290-KR-DPT-M	M	1.0	812.50		812.0

[17218 rows x 10 columns]

Check for duplicates and null values

duplicates_count = df.duplicated().sum()

null_values_count = df.isnull().sum()

duplicates_count

null_values_count

index 0

DATE 0

Months 0

CUSTOMER 0

Style 0

SKU 0

Size 0

PCS 0

RATE 0

GROSS AMT 0

dtype: int64

Clean the data

df_cleaned = df.drop_duplicates().dropna()

df

	index	DATE	Months		CUSTOMER	Style	\
0	0	2021-06-05	Jun-21	REVATHY	LOGANATHAN	MEN5004	
1	1	2021-06-05	Jun-21	REVATHY	LOGANATHAN	MEN5004	

2	2	2021-06-05	Jun-21	REVATHY	LOGANATHAN	MEN5004
3	3	2021-06-05	Jun-21	REVATHY	LOGANATHAN	MEN5009
4	4	2021-06-05	Jun-21	REVATHY	LOGANATHAN	MEN5011
...	...	...	...		...	...
17230	17230	2022-03-31	Mar-22		AVIN	SET276
17231	17231	2022-03-31	Mar-22		AVIN	PJNE3423
17232	17232	2022-03-31	Mar-22		AVIN	PJNE3404
17233	17233	2022-03-31	Mar-22		AVIN	PJNE3423
17234	17234	2022-03-31	Mar-22		AVIN	SET290

	SKU	Size	PCS	RATE	GROSS	AMT
0	MEN5004-KR-L	L	1.0	616.56		617.0
1	MEN5004-KR-XL	XL	1.0	616.56		617.0
2	MEN5004-KR-XXL	XXL	1.0	616.56		617.0
3	MEN5009-KR-L	L	1.0	616.56		617.0
4	MEN5011-KR-L	L	1.0	616.56		617.0
...						
17230	SET276-KR-PP-L	L	1.0	1425.00		1425.0
17231	PJNE3423-KR-4XL	4XL	1.0	537.50		538.0
17232	PJNE3404-KR-4XL	4XL	1.0	500.00		500.0
17233	PJNE3423-KR-4XL	4XL	1.0	537.50		538.0
17234	SET290-KR-DPT-M	M	1.0	812.50		812.0

[17218 rows x 10 columns]

Convert DATE column to datetime

```
df_cleaned['DATE'] = pd.to_datetime(df_cleaned['DATE'],
errors='coerce')
```

```
df_cleaned['DATE']
```

Convert numeric columns to appropriate types

```
df_cleaned['PCS'] = pd.to_numeric(df_cleaned['PCS'], errors='coerce')
```

```
df_cleaned['RATE'] = pd.to_numeric(df_cleaned['RATE'],
errors='coerce')
```

```
df_cleaned['GROSS AMT'] = pd.to_numeric(df_cleaned['GROSS AMT'],
errors='coerce')
```

```
df_cleaned['PCS']
```

```
df_cleaned['RATE']
```

```
df_cleaned['GROSS AMT']
```

0	617.0
1	617.0
2	617.0
3	617.0
4	617.0

...	
17230	1425.0
17231	538.0
17232	500.0
17233	538.0

```
17234      812.0
Name: GROSS AMT, Length: 17218, dtype: float64
```

```
# Show summary after cleaning
```

```
df_cleaned_info = {
    "original_shape": df.shape,
    "cleaned_shape": df_cleaned.shape,
    "duplicates_removed": duplicates_count,
    "null_values_removed": null_values_count.to_dict()
}
```

```
df_cleaned_info
```

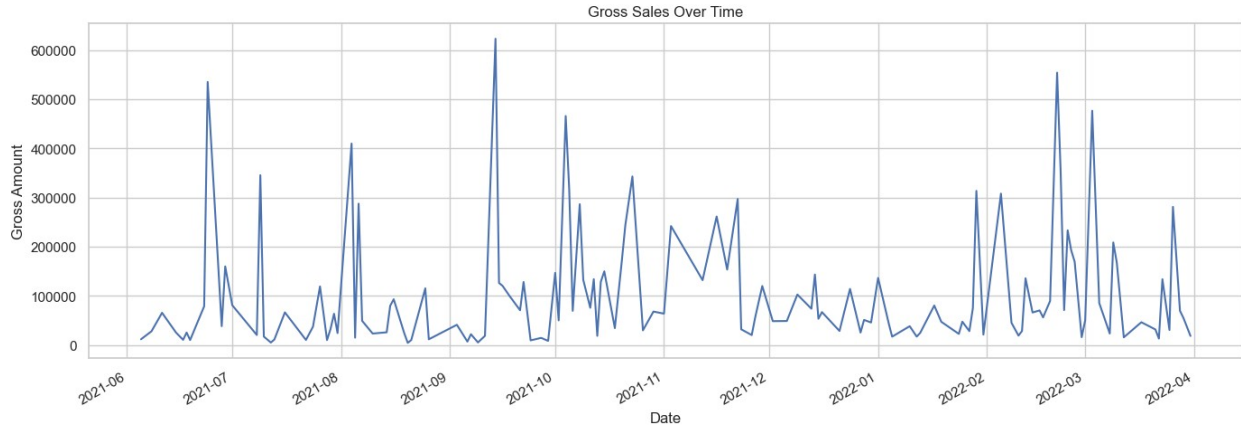
```
{'original_shape': (17218, 10),
 'cleaned_shape': (17218, 10),
 'duplicates_removed': 0,
 'null_values_removed': {'index': 0,
 'DATE': 0,
 'Months': 0,
 'CUSTOMER': 0,
 'Style': 0,
 'SKU': 0,
 'Size': 0,
 'PCS': 0,
 'RATE': 0,
 'GROSS AMT': 0}}
```

```
# Set plot style
```

```
sns.set(style="whitegrid")
```

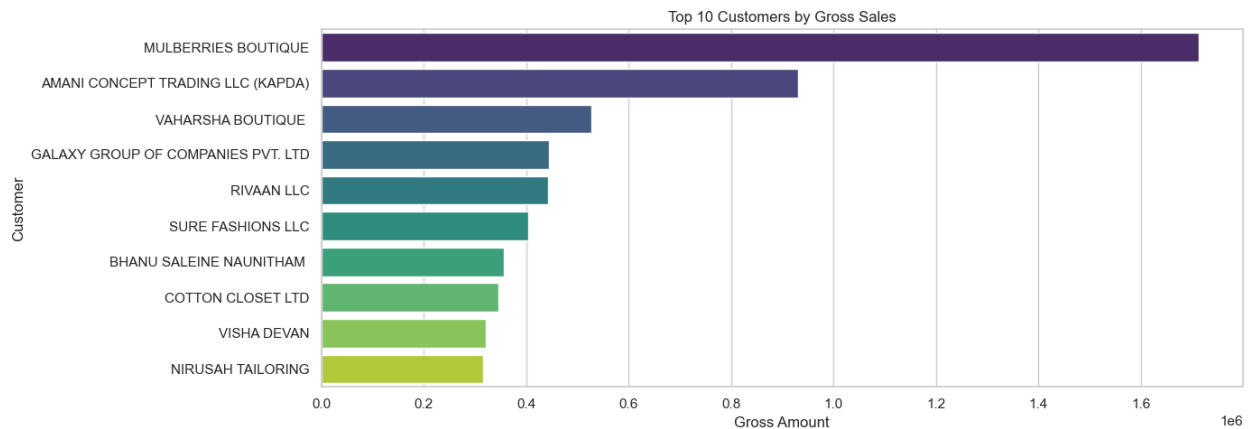
```
# 1. Sales Trend Over Time
```

```
plt.figure(figsize=(14, 5))
df.groupby('DATE')['GROSS AMT'].sum().plot()
plt.title('Gross Sales Over Time')
plt.xlabel('Date')
plt.ylabel('Gross Amount')
plt.grid(True)
plt.tight_layout()
plt.show()
```



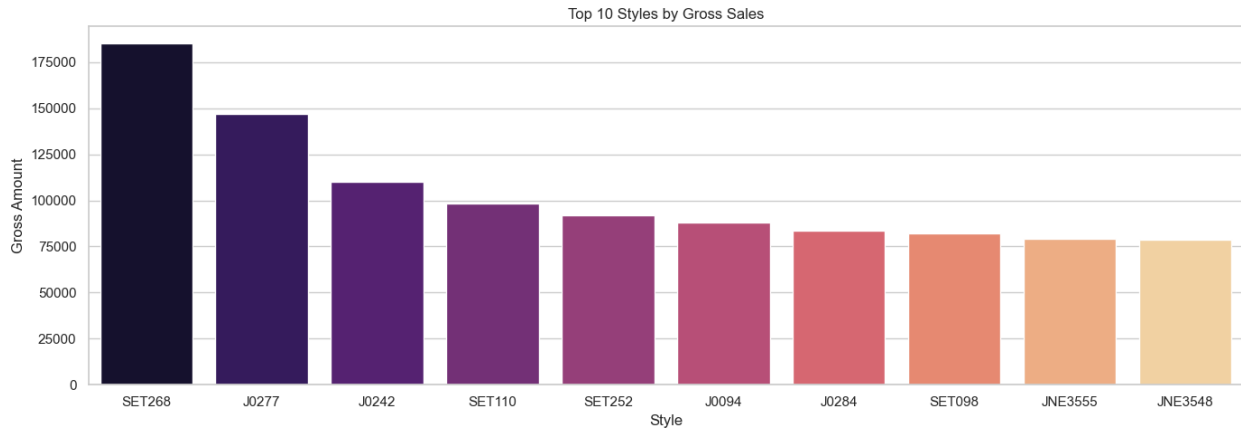
2. Top 10 Customers by Gross Sales

```
plt.figure(figsize=(14, 5))
top_customers = df.groupby('CUSTOMER')['GROSS AMT'].sum().nlargest(10)
sns.barplot(x=top_customers.values,
y=top_customers.index, hue=top_customers.index, palette="viridis",
legend=False)
plt.title('Top 10 Customers by Gross Sales')
plt.xlabel('Gross Amount')
plt.ylabel('Customer')
plt.tight_layout()
plt.show()
```



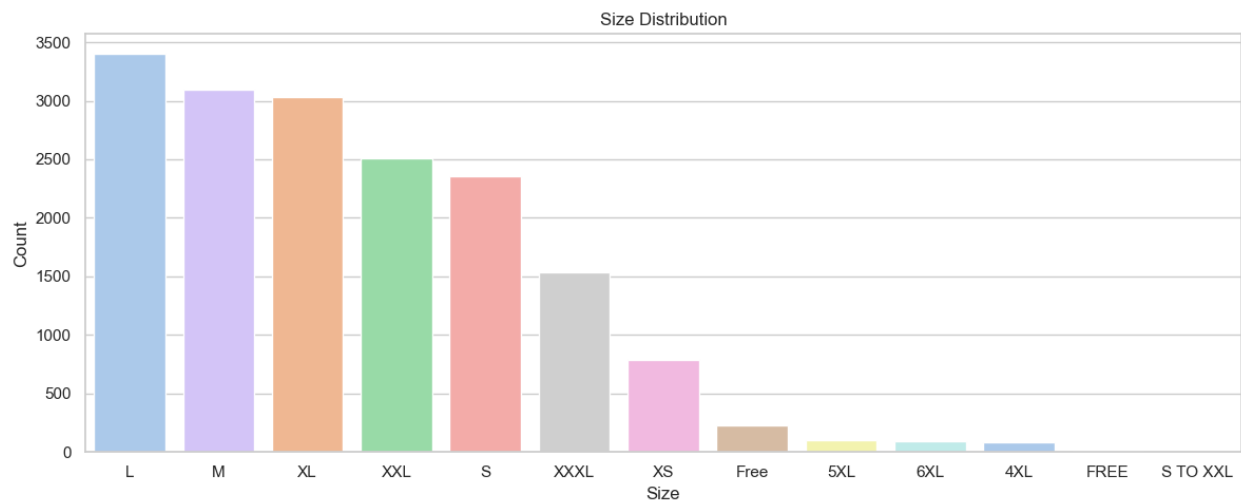
3. Style-wise Gross Sales

```
plt.figure(figsize=(14, 5))
top_styles = df.groupby('Style')['GROSS AMT'].sum().nlargest(10)
sns.barplot(x=top_styles.index, y=top_styles.values,
hue=top_styles.index, palette="magma", legend=False)
plt.title('Top 10 Styles by Gross Sales')
plt.xlabel('Style')
plt.ylabel('Gross Amount')
plt.tight_layout()
plt.show()
```



4. Size Distribution

```
plt.figure(figsize=(12, 5))
sns.countplot(data=df, x='Size',
order=df['Size'].value_counts().index, hue='Size', palette="pastel",
legend=False)
plt.title('Size Distribution')
plt.xlabel('Size')
plt.ylabel('Count')
plt.tight_layout()
plt.show()
```



5. PCS vs Gross Amount

```
plt.figure(figsize=(12, 5))
sns.scatterplot(data=df, x='PCS', y='GROSS AMT', hue='Size',
alpha=0.6)
plt.title('PCS vs Gross Amount by Size')
plt.xlabel('PCS')
plt.ylabel('Gross Amount')
plt.legend(title='Size', bbox_to_anchor=(1.05, 1), loc='upper left')
```

```
plt.tight_layout()
plt.show()
```

